Space-borne four-face AESA radar

Design and modelling reference of the mission-preparation tool

C.44 project

25 October 2026 – radar model V6.16.0 (roadmap step 4.1 coded)

Abstract

This document describes, for a reader who knows radar but not this project, the radar that the mission-preparation tool models, the radio-frequency (RF) and signal-processing models it implements, and the design choices behind every value. The radar is a four-face, fully digital active electronically scanned array (AESA) carried by a spacecraft, using S band for search and Ku band for confirmation and tracking. The tool lets an operator describe a mission (watch volumes, threats, sky, navigation) and validates it against the resources of the whole system: RF budget, power, time on air, environment, and, in later steps, scheduling, thermal and computing loads. The document follows the code and the single parameter file of the repository; where the two differ, the parameter file is the reference.

Read this first. (1) The test missions of the repository and their frozen outputs are *test values*: they exist only to detect regressions of the code. They are neither requirements nor demonstrated performances of the radar. (2) Technology values (TRMs, receivers, converters, oscillator) are *credible but optimistic assumptions for the 2050–2060 time frame*: when a range of values was considered, its optimistic end was kept. Threat and environment values are *class means* or medians, not data on a specific system. (3) All values live in one file, `radar_system.toml`; each entry carries its value, unit, status and the reason for the value. This document explains those reasons; it does not replace the file.

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1 Purpose and scope

1.1 What the tool does

The tool is a *mission-preparation* tool. The operator describes what the radar must do during a mission:

- **watch volumes** ("sectors"): an angular box or cap, a required detection range, a revisit period, a threat (class of target, radar cross-section, speed), resolution and false-alarm settings, optional active-tracking range brackets;
- the **sky**: bodies in front of the faces (Sun, Earth, Moon, ...), a galactic frame, stand-off jammers;
- the **navigation**: the attitude of the spacecraft and, optionally, its orbit;
- global settings: optional power cap per face (the faces share the ship's 8 MW), spectrum the radar must not use, elevation window.

For each volume, the tool chooses the S-band search frequency and waveforms, computes the achieved ranges and the energy and time each face spends, adds the Ku-band confirmation of every detection and the active tracking, and flags whatever the system cannot do (power or time over-subscription, unreachable range, blinded directions, scan limits, self-protection jamming, ...). The aim is to validate the mission *against the whole radar system*, not only against a link budget.

1.2 Place in a larger simulation

The tool is the *radar computer* layer of a four-layer architecture (Figure 1) that separates *truth*, known only to the simulation, from *knowledge*, what the spacecraft knows:

1. **global scenario simulation**: orbits, objects, Newtonian propagation (the truth);
2. **converter**: turns the truth into the spacecraft frame for the RF simulation, and into a navigation message for the radar computer;
3. **radar computer** (this tool): knows only its navigation and its own detections; keeps the volumes attached to the spacecraft, schedules, chooses waveforms;
4. **RF simulation**: plays the orders of the radar computer against the truth.

In mission preparation, the navigation is a snapshot entered by the operator; later it will be the navigation message of the global simulator. Volumes and sky are given in a *mission frame* (ship, inertial, RTN, limb, two directions); the radar computer applies the rotation by the attitude (Section 15).

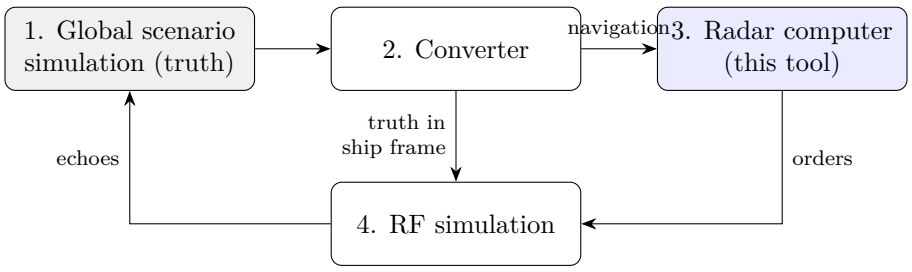


Figure 1: Four-layer architecture: truth and knowledge are kept apart. The tool implements layer 3 in mission preparation.

1.3 Design method

The model was rebuilt step by step along a roadmap (Section 17). A few rules shape every part of it:

- **One parameter file.** Every system value is an entry of `radar_system.toml` with `value`, `unit`, `status` (V: locked by the designer; D: derived or proposed, to confirm), `source` (the reason) and `date`. Tool settings (search grids, caches, defaults) are in a separate `[tool.*]` block. 290 system entries exist today, 247 of them locked.
- **Derived, not entered.** Anything that follows from other values is computed: the array build-up, the radiation patterns, the system noise temperature, the converter noise figure, the longest pulse, the frequency grid, ... The loader refuses inconsistent combinations with a clear message.
- **Optimistic technology, median threats.** For our technology, the optimistic end of a credible 2050–2060 range is kept; for threats and the environment, the median or class mean.
- **Approximations by decision only.** Exact shortcuts (closed forms, caches, exact pruning) come first; since 25 October 2026 approximations are allowed when they do not denature the model, each one decided by the designer with its measured effect and kept as a setting whose exact value stays available.
- **Test values are not performances.** The test missions only guard against regressions.

2 The radar

2.1 Faces and geometry

The radar has **four planar faces** spread at 90° around the spacecraft axis (azimuths 0°, 90°, 180° and 270°), all **tilted by 25° toward the nose**. The ship frame has Z along the axis, toward the nose ("lighthouse" model: the axis is vertical, the nose at the zenith); azimuth φ is counted about Z from $+X$ toward $+Y$, elevation ε from the XY plane toward $+Z$.

Each face covers a *sector* of radius 75° around its normal (the widest scan limit of the bands), cut by the planes $\varphi = \pm 45°$ between neighbouring faces. Search positions are counted in the direction cosines (u, v) of the face: a planar array scans with a beam of constant width in (u, v), which widens as $1/\cos\theta$ in angle. The sector covers an area of 2.0006 in (u, v).

The spacecraft geometry comes from a low-polygon 3D model (149.46 m long, 74.68 m span over its radiator wings, hull diameter 17.22 m); the four radar panels are octagons sitting on a cone around the nose station, with the aperture centres 5.248 m from the axis. The loader checks the model against the locked dimensions and checks that each aperture centre lies on a panel whose normal is the face normal at the locked tilt.

2.2 Bands and roles

	S band	**Ku band**
Usable band	2.60 GHz to 2.93 GHz	13.5 GHz to 14.5 GHz
Core band (no edge loss)	2.66 GHz to 2.88 GHz	13.7 GHz to 14.3 GHz
Role	search (detection of new targets)	confirmation of detections, active tracking
Scan limit	75° up to 2.85 GHz, 70° above	75° up to 14.25 GHz, 70° above
TRMs per face	4 900	122 500
Peak power per TRM (PA output)	2 000 W	75 W
Peak power per face	9.80 MW	9.19 MW
Instantaneous bandwidth	200 MHz	200 MHz
Cold-sky system temperature	144.5 K	230.8 K

The architecture is **fully digital**: one analogue-to-digital converter per TRM and per polarisation channel, digital true time delay in every TRM (no beam squint, no aperture-fill limit on the bandwidth), receive beams formed by computation. Search uses S band only; the S "touch" of one confirmation mode is the single exception.

2.3 Array build-up

Each face is a dual-band tiling of **196 TRSAs** (transmit-receive sub-arrays). A TRSA is a square tile of 25×25 Ku TRMs (pitch 10.4 mm) and 5×5 S TRMs (pitch 52 mm, close to $\lambda/2$ in S); each S TRM covers exactly one block of 5×5 Ku TRMs. The TRSAs are laid out in an approximate octagon of 16 centred rows,

$$[6, 8, 10, 12, 14, 16, 16, 16, 16, 16, 16, 14, 12, 10, 8, 6],$$

separated by a mechanical gap of 0.5 mm (a dead, non-radiating region). Derived quantities:

TRSA side	260 mm
Emissive area	13.249 6 m^2 (fill 99.65 % of 13.296 4 m^2)
Envelope D_{env} (widest row: 16 TRSAs and 15 gaps)	4.167 5 m
Mean element pitch (TRSA pitch / TRMs per side)	S 52.1 mm, Ku 10.42 mm
RMS extent of the aperture along one axis x_{rms}	1.033 m = 0.248 D_{env}

The gap was reduced from 2 mm to 0.5 mm because the break of the element pitch from one TRSA to the next creates "gap lobes" near the grating lobe (−16.6 dB in Ku with 2 mm) that no receive weighting can lower; 0.5 mm assumes TRSA and back-plate materials matched in thermal expansion.

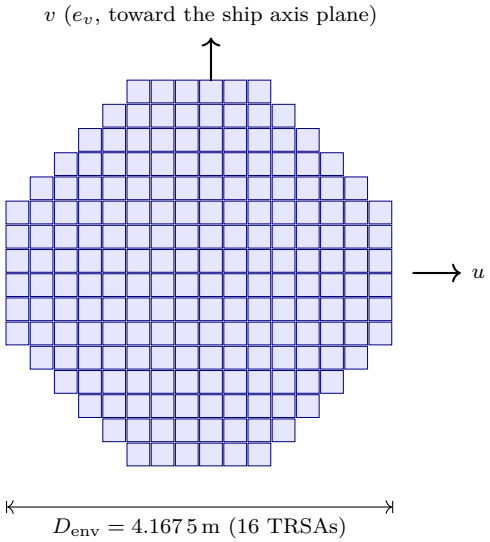


Figure 2: The 196 TRSAs of one face (each square: 25×25 Ku TRMs and 5×5 S TRMs).

3 Antenna model

3.1 Radiation pattern computed from the build-up

The pattern is not a textbook formula: it is computed from the element positions of the build-up (`core/antenna.py`). Elements are isotropic in the array factor; the element pattern enters only through the projected area $\cos\theta$ and the active mismatch (Section 3.3). Near boresight, expressed

in units of λ/D_{env}, the pattern does not depend on frequency, so one computation per band serves the whole band (S and Ku sample the same octagon and agree to 0.1 %). The rows are centred and palindromic, so the array factor is real,

$$AF(u,v) = \sum_{\text{quadrant}} w\, m\, \cos(kxu)\cos(kyv),$$

with multiplicities m for the symmetry; with the $x \leftrightarrow y$ symmetry of the octagon, cuts from 0° to 45° suffice, each a matrix product. Results are cached under a hash of their inputs.

- **Transmit beam**: uniform illumination (PAs saturated). Half-power beamwidth (smallest cut) $1.036\,\lambda/D_{\mathrm{env}}$, i.e. 1.525° at 2.8 GHz and 0.305° at 14 GHz; first sidelobe −15.6 dB.
- **Receive beam**: circular Taylor (Hansen) law, $\bar{n} = 5$, whose design level is *fitted numerically* (regula falsi) so that the level *achieved* on the real octagon equals the option chosen for the watch volume: −25 dB, −30 dB and −35 dB. Efficiency and beamwidth follow:

Option	design level	η_{RX}	HPBW (λ/D_{env})
−25 dB (default)	29.5 dB	0.873	1.173
−30 dB	35.7 dB	0.773	1.255
−35 dB	41.7 dB	0.693	1.325

  Lowering the level rejects strong off-axis sources (bright target, Moon or Earth, jammer) at the cost of receive gain (−0.53 and −1.00 dB against the −25 dB option). One setting per volume serves S and Ku.
- **Gain**: $G = 4\pi A\,\eta\,\eta_{\mathrm{ohm}}\cos\theta/\lambda^2$ with the emissive area A; at 2.8 GHz on boresight, 41.62 dBi in transmission and 41.03 dBi in reception (−25 dB); Ku transmit gain at 14 GHz: 55.60 dBi.
- **Angle accuracy**: the Cramér–Rao bound of the full digital aperture with a margin of 1.13,

$$\sigma_u = 1.13\,\frac{\lambda}{2\pi\, x_{\mathrm{rms}}\sqrt{2\,\mathrm{SNR}}},$$

  independent of the receive weighting (the maximum-likelihood estimator uses the whole aperture); the same margin applies to the Doppler and range estimators.

3.2 Far sidelobes and gap lobes

Beyond the near window ($8\,\lambda/D_{\mathrm{env}}$), the array factor is scanned on a 2D grid over the whole visible region of the band (radius $(1+\sin\theta_{\mathrm{lim}})D_{\mathrm{env}}/\lambda$ at the top of the band), grating-lobe neighbourhoods excluded. A cumulative profile stores the peak lobe level within a distance r of the beam axis; a lobe at r becomes visible once the beam is scanned to θ with $r\lambda/D_{\mathrm{env}} \le 1 + \sin\theta$ (worst azimuth). The strongest far lobes, gap lobes included, are −25.3 dB in Ku and −30.7 dB in S (uniform); weighting does not lower gap lobes. For each sector the tool reports up to which scan angle the receive option is held over the visible region; in Ku the −30 dB and −35 dB options are held up to 75° and 62° at 14 GHz only.

The same module provides *azimuthal means* of the receive power pattern versus the distance to the beam axis (fine near the axis, annulus means beyond). They are used for the sky-noise integral, the sidelobe clutter and the jammer pick-up (Sections 6.5 and 11). By Parseval's theorem the normalisation is exact: a body filling the whole front half-space adds its brightness temperature (checked to within 1.00–1.04).

3.3 Active mismatch and grating lobes

One model, applied to S and Ku, in transmission and in reception, describes the active reflection of the array: a current sheet matched at $\theta_m = 45°$ by a wide-angle impedance-matching (WAIM) layer, and the rise of the reflection as the first grating lobe approaches the visible region ($g_0 = 0.02$, d_{mean} the mean element pitch, worst azimuth):

$$|\Gamma_{\text{WAIM}}| = \frac{|\cos\theta - \cos\theta_m|}{\cos\theta + \cos\theta_m}, \qquad |\Gamma_{\text{GL}}| = \exp(-g/g_0), \quad g = \frac{\lambda}{d_{\text{mean}}} - \sin\theta - 1,$$
$$1 - |\Gamma|^2 = (1 - |\Gamma_{\text{WAIM}}|^2)(1 - |\Gamma_{\text{GL}}|^2),$$

with a loss per pass $-10\log_{10}(1 - |\Gamma|^2)$, counted twice (transmit and receive). At 2.8 GHz the round trip costs 0.26 dB on boresight, 0 at 45°, 1.12 dB at 70° and 2.11 dB at 75°. The upper edge of the S band is set by this wall: at 2.93 GHz and 70° the loss is 0.97 dB per pass; a grating lobe becomes visible at 3.0 GHz. The optimiser flags any bin used beyond 3 dB per pass.

The transmit polarisation is linear along e_v (the face axis in the plane of the ship axis), one PA per TRM; its purity (Ludwig-3 cross-polarisation, $(1 + \cos\theta)/(1 - \cos\theta)$ in the 45° plane) is reported. With the dual-polarisation receiver it costs nothing in the budget.

3.4 Sub-apertures for the transmit partition

For the aperture-partition multibeam mode (Section 7.4) the face is cut into halves (horizontal cut `h`: rows above the centre; vertical cut `v`: columns right of the centre) or quadrants, each uniform. Their 2D patterns are evaluated directly (a half is not centred, so no symmetry is assumed), exactly, by a sum separable by element rows:

Aperture	S TRMs	-3 dB widths $u \times v$ (λ/D_{env})	lattice cell / full face
Full face	4 900	1.041×1.041	1
Half	2 450	1.041×1.928	$\times 1.85$
Quadrant	1 225	1.928×1.928	$\times 3.43$

The cell ratio is below 2 and 4 because half an octagon is not half as wide in effective width. Because the octagon is symmetric in x and y, the two halves are the same beam rotated by 90°: the code checks this exactly and then searches only one of them.

4 Transmit chain and power budget

4.1 TRM model

The TRMs are GaN-on-diamond with micro-channel cooling. Peak power is specified at the PA output (where the power-added efficiency, PAE, is defined); the path from the PA to the element is a derived loss (harmonic filter, circulator, calibration coupler, line: 0.6 dB in S, 0.95 dB in Ku).

PAE. For a pulse of length τ in a train of duty cycle $d = \tau/\text{PRI}$,

$$\eta = \eta_{\text{sat}} \frac{\tau}{\tau + t_{\text{edge}}} \min(1,\ 1 - a\,(d - d_{\text{ref}})),$$

with the saturated PAE η_{sat} at the reference duty $d_{\text{ref}} = 0.10$, the time t_{edge} during which the PA draws power without full RF (switching, bias and phase settling), and a slope a for channel heating. A single pulse uses d_{ref}.

	S	Ku
η_{sat}	0.80	0.65
t_{edge}	0.2 µs	0.1 µs
Duty slope a	0.05 ($-2.75\,\%$ at $d = 0.65$)	0.08 ($-4.4\,\%$ at $d = 0.65$)
Duty ceiling on trains	0.65	0.65
Energy buffer per TRM	25 J	1.2 J
Longest pulse $\tau_{\max}$	2.0 ms	2.08 ms

Energy buffer. Each TRM draws its pulse from a local capacitor bank. A pulse using a fraction $x = E/E_{\text{buf}}$ of it loses on average $-10\log_{10}(1 - x/2)$ (the PA output follows the stored energy); a pulse beyond $x = 0.2$ is refused, which sets $\tau_{\max} = 0.2\,E_{\text{buf}}\,\eta_{\text{sat}}/P_{\text{pk}} - t_{\text{edge}}$. The longest pulse ends 0.97 dB down (mean 0.46 dB); the AM/PM conversion of the PA as the buffer empties is reported, not counted (quasi-linear phase drift).

4.2 Electrical budget

The budget of a face is *electrical*: every look costs its RF energy divided by the PAE of its pulse, and the heat of the face is the electrical energy minus the RF energy (about 370 W per S TRM during a long train at $d = 0.65$, an input of the thermal step). Since roadmap step 4.1 the electrical power is modelled at four levels.

Ship level: a shared budget. The spacecraft provides 8 MW to the radar. The faces do *not* have fixed shares: they draw from this budget in common, so a heavily loaded face may take more than a quarter when the others need less. The verdict on energy is therefore taken for the ship,

$$\sum_{k=1}^{4} P_{\text{bus},k} \le 8\,\text{MW}, \qquad P_{\text{bus},k} = \frac{P_{\text{TX},k} + P_{\text{on}} + p_{\text{pump}}\,Q_k}{\eta_{\text{psu}}\,\eta_{\text{dist}}},$$

where $P_{\text{TX},k}$ is the mean electrical power of the transmissions of face k (search, confirmation, tracking, at the TRM supply), P_{on} the always-on consumption, Q_k the mean heat of the face and $p_{\text{pump}} = 2\,\%$ the share of that heat spent by the cooling pumps; $\eta_{\text{psu}} = 0.95$ and $\eta_{\text{dist}} = 0.995$ are the efficiencies of the power supply and of the distribution (product 0.945). Each face is also limited by what its own feeder (cables, converter) can carry, 8 MW, and by an optional cap the operator may reserve for it in the mission (one value for all faces or one per face). The tool reports the power each face actually draws, the margin left in the shared budget, and a suggested split proportional to the needs. To weigh energy against time when it chooses waveforms, the search optimiser keeps a *reference* allocation of 8 MW/4 = 2 MW per face; the joint optimisation of the split belongs to the global optimiser (step 5.2).

Always-on consumption. A digital array receives all the time, whether it transmits or not. Per face: two converters per TRM (one per polarisation; 0.1 W each at 250 MS/s and 13 bits, an optimistic 2050 value against 0.3–1 W today) and two receive chains per TRM (LNA, amplifiers, local-oscillator distribution: 0.5 W in S, 0.1 W in Ku), in both bands; 50 kW of digital beamforming and processing (provisional, until the computing step); 2 kW of control. Total 106.9 kW per face (9 800 S and 245 000 Ku converters). With the reference 2 MW, a face has $2 \times 0.945 - 0.107 = 1.784$ MW left for its transmissions.

TRM level: supply over a train. Each TRM draws its pulse from its local buffer and refills it from a continuous supply (1 700 W in S, 80 W in Ku). Over a train of n pulses of energy $E_p = P_{\text{pk}}\tau/\eta$ every PRI, two conditions must hold: the mean power E_p/PRI stays within the

supply, and the buffer, recharged at the supply rate between pulses, loses at most the allowed fraction:

$$(n-1)\,\max\bigl(0,\ E_p - P_{\text{sup}}\,\text{PRI}\bigr) + E_p \le 0.2\,E_{\text{buf}}.$$

A waveform that breaks either condition is removed from the search spaces (no S waveform today; 68 of 2 356 Ku looks, long trains at high duty).

Face level: storage and peak power. While a face transmits, all its TRMs draw together: about 12.3 MW for a full-power S look ($4\,900 \times 2\,\text{kW}/0.8$) and 14.1 MW in Ku, far above the mean the bus delivers. The difference comes from an energy storage per face (supercapacitors, 5 MJ, 80 % usable, 15 MW discharge), recharged at the rate the feeder leaves after the always-on consumption. Three checks:

- *one observation*: the energy of the costliest observation of the plan (an S search dwell or a Ku look), minus the recharge during it, must fit in the usable storage;
- *worst burst*: if all the transmissions of the longest revisit of the face were packed together, the deficit $RR\,(P_{\text{TX}} - P_{\text{rech}}\,f_t)$ (f_t the share of time on air) must fit in the usable storage; it never does for long revisits, so this check yields a *scheduler rule*: at most $E_{\text{usable}}/(P_{\text{peak}} - P_{\text{rech}})$ of contiguous transmission at peak power (about 0.3 s), then recharge;
- *peak power*: the peak of an observation must stay within the storage discharge plus the recharge.

The step of the bus load at the start of an observation and its slope over the pulse edge (0.2 ţs in S) are reported for the design of the bus; they are not modelled electrically.

5 Band model

Each band has a usable band and a core band. A signal of bandwidth B at centre f occupies $[f - B/2,\ f + B/2]$, which must lie inside the usable band (and outside the spectrum the mission forbids); otherwise the waveform is not playable at f. Outside the core band, a raised-cosine derating reaches `edge_derating_db` at the usable edges (1.0 dB round trip in S, 0.5 dB in Ku); its *exact* average over the occupied spectrum is applied. The loader checks that the core band and the scan-limit break frequency lie inside the usable band, so the band can be changed in the file alone. With digital true time delay there is no aperture-fill limit (phase steering would limit the signal to $c/(D_{\text{env}} \sin\theta)$, about 75 MHz at 75°).

6 Receive chain

6.1 System noise temperature

T_{sys} is the Friis cascade referred to the antenna port: cold-sky antenna temperature, the losses before the LNA (circulator, limiter, line: they attenuate and add $(L-1)T_{\text{phys}}$), the LNA, the post-LNA chain and the ADC, each divided by the gain ahead of it, and a digital-noise factor. Values (S / Ku): cold sky 3.2 K / 2.8 K, front-end at 320 K, pre-LNA losses 0.85 dB / 1.2 dB, LNA noise figure 0.75 dB / 1.15 dB, giving **144.5 K in S and 230.8 K in Ku** for the cold sky. The sky in front of the face is added per direction (Section 6.5).

6.2 Converter and dynamic range

The ADC noise figure is derived from the converter, not entered:

$$\text{NF}_{\text{ADC}} = P_{\text{FS}} - (6.02\,\text{ENOB} + 1.76) - 10\log_{10}(kT_0 f_s),$$

with 13 effective bits, a full scale of 10 dBm and $f_s = 1.25 \times 200\,\text{MHz} = 250\,\text{MS/s}$ complex: 20.0 dB. Levels are referred to the antenna port: the clipping level is $P_{\text{FS}} - 3\,\text{dB}$ minus the gain to the ADC (53.4 dB above the noise in S). Digitisation happens per element, *before* beamforming: a strong source in any front direction reaches every converter. The *minimum unsaturated range* is where the peak echo per element of a full-power, single-beam look reaches the clipping level (S: 12.2 km for the reference target, 2.3 km for a KKV). The receiver is deaf for τ plus the limiter recovery (50 ns in S, 25 ns in Ku).

6.3 Dual polarisation

Each TRM has two receive channels. A detection look uses a GLRT detector on the unknown polarisation of the echo, estimated over the look: one more unknown complex sample, a loss $\text{SNR}_{\text{req}}(n+1) - \text{SNR}_{\text{req}}(n)$ (0.47 dB for one coherent sample, small for long noncoherent trains). A tracking look knows the polarisation from the track (0.05 dB residual). The dual-channel receiver also collects the cross-polar echo of the target, a gain $10\log_{10}(1+\rho)$ with ρ the cross/co-polar RCS ratio, and the two channels add polarisation diversity to the target fluctuation (Section 10).

6.4 Isolation between bands and faces

Leakage from each transmitting band and face to each receiving element is checked against three limits: LNA blocking, ADC clipping and in-channel desensitisation (6 dB below the noise). With the couplings assumed (90 dB to an adjacent face, 120 dB to the opposite one, −30 dB inside a TRSA) a face cannot receive a band while another face transmits it: all faces share **a common transmit/receive cadence per band**. A further rule: no Ku reception on a face during an S transmission whose fifth harmonic $[5(f - B/2),\ 5(f + B/2)]$ covers the Ku channel. These are rules for the scheduler (not yet enforced; no loss is put in the budget before the scheduling step).

6.5 Sky noise

The antenna temperature depends on the look direction. With the receive gain written $G = G_0\, w\, g(\rho)$ (element pattern $w = \cos\theta$, nothing behind the face) and $d\Omega = du\, dv/w$, the factor w cancels and the rise is

$$\Delta T = T_b \,\frac{A\,\eta_{\text{RX}}\,\eta_{\text{ohm}}}{D_{\text{env}}^2} \int_{\text{body}} g(\rho)\, d\rho^2,$$

the integral running over the body as seen in the (u, v) plane of the face (front half only), ρ being the distance to the beam axis in λ/D_{env} and g the azimuthal mean of the computed receive pattern. The main lobe is therefore exact, not a Gaussian approximation. Small bodies (Sun, Moon) are sampled finely on their disk; large ones (Earth) by a polar grid in the near window and rings beyond. Brightness temperatures are class values per band (quiet, mid-cycle and active Sun: 4.5×10^4 K, 1.0×10^5 K and 1.5×10^5 K in S; Earth 290 K worst case, ocean and land variants; Moon 230 K); the galactic background follows a plane-and-bulge model scaled as $f^{-2.5}$, with a worst case applied everywhere when the mission gives no galactic frame. The Sun in the beam adds about 17 dB in S as in Ku; the Moon about 2.6 dB in Ku.

For a sector, T_{sys} is evaluated every half transmit beamwidth over the scanned positions (raster of 0.2°). Near hot sources the noise varies faster than that step, so the raster cells are evaluated one by one where the rise of their block exceeds 0.5 dB, and within three transmit beamwidths of a stand-off jammer, of a small body's centre or of a large body's limb; each such cell is then split into 8×8 points of equal weight. A point whose rise exceeds 10 dB is *blinded*: the cell is searched only for the share of its points left, and the blinded share is reported with the body responsible. The *budget* T_{sys} of the sector is the $(1 - \texttt{short_tol})$ quantile (99 %) of the positions not blinded, every point counting for its own weight, and all its ranges scale with $(T_{\text{cold}}/T_{\text{budget}})^{1/4}$. Turning the sky and the sector together about the ship axis by angles off the raster moves the budget and the blinded share by less than 1 % (Sun, 0.37° and 1.13°); a steady drift at larger angles (1.4 % at

2.9°), believed to come from the fixed faces, is still to be confirmed. The same rule applies in Ku to the confirmation and tracking of the sector.

7 Waveforms

7.1 Modulation and pulse compression

Two families are modelled: linear FM (Doppler tolerant) and non-linear FM (low range sidelobes shaped by the frequency law, no weighting loss); phase codes are left out because they tolerate a Doppler shift poorly at closing speeds up to 25 km/s ($f_d \approx 470$ kHz in S). For each look, with $f_{d,\max} = 2v/\lambda$:

- if $f_{d,\max}/B \leq 2\,\%$: NLFM, mismatch loss 0.1 dB;
- otherwise: LFM with a Taylor ($\bar{n} = 5$) receive-only weighting at the sidelobe level of the volume; loss = weighting loss $- 20\log_{10}(1 - f_d/B)$ (Doppler loss at the peak of the ambiguity function).

The range sidelobe level is the same as the angular option of the volume: one sidelobe policy per volume. Weighting at −25 dB, −30 dB and −35 dB costs 0.41 dB, 0.68 dB and 0.93 dB and widens the main lobe by ×1.18, 1.27, 1.34. The physical range resolution is this widening times $c/2B$ (2 m requires 89 MHz at −25 dB); the range estimator constant is $k_r = 0.6/$(RMS-bandwidth ratio). The range–Doppler coupling bias $cf_d\tau/2B$ is corrected by the Doppler measurement.

7.2 Intra-pulse effects and timeline

A target crosses $n_c = 2v\tau B/c$ range cells during the pulse; a filter matched to another speed then has a quadratic phase error, a loss $-20\log_{10}\left|\langle e^{j2\pi n_c x^2}\rangle_{x\in[-1/2,1/2]}\right|$ (1 dB at $n_c = 1.016$). This sets the spacing of the velocity bank (Section 8).

The receiver is deaf during the pulse and the limiter recovery: blind range $c(\tau + t_{\text{rec}})/2$ (3 km for 20 µs, 300 km for 2 ms), which must stay below the sector's shortest range of interest (`range_min_km`, 10 % of the required range by default). A train is unambiguous in range when PRI $\geq 2R/c + \tau + t_{\text{rec}}$; a sector may allow ambiguous trains, staggered over 3 PRIs at ±10 %, with the mean eclipsing loss $-10\log_{10}(1 - (\tau + t_{\text{rec}})/\text{PRI})$ (they are removed when a surface is in the main lobe, since its clutter would fold onto the targets).

7.3 Frequency agility

A noncoherent train may hop its carrier pulse to pulse over an excursion Δf; the occupied band becomes $B + \Delta f$ and the number of independent target samples is $\min(n, \lfloor \Delta f/(c/2L) \rfloor + 1)$ for a target of length L, which turns a declared Swerling 1/3 target into a 2/4-like one over the train.

7.4 Transmit multibeam

In S, one dwell may serve B_{tx} beams. The digital receiver already forms many receive beams per position (no loss; a computing load). Three transmit modes are modelled:

- **Phase-only superposition** ($B_{\text{tx}} = 2, 4, 8$): every TRM sends the constant-envelope sum of the beams (PAs saturated, PAE unchanged), power P/B_{tx} per beam, full transmit gain, a loss of 0.4 dB, 0.6 dB and 0.8 dB (power lost to the intermodulation of the constant-envelope sum).
- **Aperture partition** ($B_{\text{tx}} = 2, 4$): halves or quadrants (Section 3.4), power P/B_{tx} and gain $G\,N_{\text{sub}}/N$ per beam, a wider beam on its own lattice: fewer positions.
- **Time multiplex** ($B_{\text{tx}} = 2, 4, 8$): full-power pulses sent back to back in the PRI; blind range, buffer, duty and PAE computed on $B_{\text{tx}}\tau$. Modelled, but not offered to the optimiser: in the

budget of transmit time with interleaved positions it spends the same energy and time per position as one beam and is never better. Its value is in the timeline (latency, shared echo waiting), to be re-evaluated with the scheduler.

Linear superposition (PA back-off) and frequency multiplexing were left out.

7.5 Search waveform space

The S search space is the product of: integration (coherent, noncoherent), transmit mode and beams, signal bandwidths 10 MHz, 20 MHz, 30 MHz, 50 MHz, 75 MHz, 100 MHz, 150 MHz and 200 MHz meeting the sector's resolution, pulses from 5 µs to 2 ms (13 values, the last one set by the buffer), 1 to 32 pulses, the agile variant of noncoherent trains and, where a surface clutters the sector, Taylor Doppler weightings of coherent trains. The PRI is the shortest unambiguous one (or 1 ms for single pulses and allowed ambiguous trains). About 6 500 candidates per sector.

8 Signal processing

8.1 Coherent blocks and coherence

A coherent train of N pulses is cut into M blocks (M divides N) integrated noncoherently, M minimising the required SNR of M blocks plus the coherence loss of one block. Noncoherent trains, and declared Swerling 2/4 cases, have $M = N$. The coherence loss of a block of duration T combines

- the target decorrelation $\eta = \exp(-(T/T_0)^2)$, with $T_0 = \lambda/(2L\omega)$ from the target class (length L, rotation rate ω);
- the residual phase variance σ^2, giving $4.34\,\sigma^2$ dB: oscillator phase noise filtered by the round trip, $4\sin^2(\pi f \tau_{rt})$, integrated from $1/T$ to $B/2$ (one oscillator for transmit and receive; an optimistic photonic oscillator, -150 dBc/Hz at 1 kHz from 10 GHz), plus the residual platform motion $(4\pi\delta/\lambda)^2$ with $\delta = 10$ µm rms.

The losses are negligible with these values; the visible effect is the Doppler floor, reported per look.

8.2 Velocity and acceleration banks

Range migration is compensated (keystone or Radon type, residual 0.1 dB when the target crosses more than 1.5 cells). The filter for a velocity offset Δv loses the intra-pulse term plus the range walk over the look; the bank step is chosen so that a target half a step off loses 1 dB; the *mean* loss over the step goes into the budget and the number of hypotheses into the false-alarm count (and, later, the computing load). Accelerations up to the class maximum add a quadratic phase on a coherent block and a residual range curvature; when they cost more than 1 dB, a bank of acceleration hypotheses multiplies the velocity hypotheses.

8.3 Grids, CFAR and false alarms

The detection grid is oversampled ×4 in range and the Doppler transform zero-padded ×4: mean straddle 0.05 dB and 0.07 dB. The CA-CFAR uses 1 024 reference cells (loss 0.04 dB, exact for Swerling 1 on one sample); OS-CFAR (rank 0.75) and GO-CFAR (greatest of the two half windows, which holds the P_{fa} at a clutter edge, 0.05 dB) are options. Independent cells per position: range extent over $c/2B$, Doppler cells of a block, velocity and acceleration hypotheses, and angular cells (receive beams per position × the receive-beam spacing squared). The oversampled grids are not counted (correlated samples).

9 Detection criterion

9.1 Exact detection laws

A square-law detector on n noncoherently integrated samples (the coherent blocks) is modelled exactly: under noise alone the statistic is Gamma$(n, 1)$, so $P_{\mathrm{fa}} = P(\mathrm{Poisson}(T) \le n - 1)$; for a steady target of total SNR λ, $P_{\mathrm{d}} = \sum_k \mathrm{Pois}(k; \lambda)\, P(\mathrm{Poisson}(T) \le n + k - 1)$. Fluctuating targets are mixtures over a Gamma-distributed target power of shape k ($k = 1$: Swerling 1, 2: Swerling 3, n: Swerling 2, $2n$: Swerling 4, ∞: steady); non-integer shapes are allowed, which is how decorrelation inside a look and polarisation diversity enter. The laws are valid for any P_{fa} and agree with Shnidman's fit within its 0.3 dB error.

9.2 Pd averaged over the target position

The random position of the target costs SNR. The offsets in the hexagonal transmit cell (real pattern), in the receive beam lattice, in range and Doppler straddle, and to the nearest velocity filter form a discrete loss law (16 quantiles each, combined as independent). The P_{d} of a look is the *mean* P_{d} over this law, and the SNR it costs is the budget loss called "positions". A spacing of 1.4 transmit beamwidths is the optimum of this criterion for one look (flat within 0.1 dB from 1.3 to 1.6); since the cumulative criterion lowers the P_{d} asked of a look, the optimiser chooses the spacing of each sector among 1.3 to 2.0 by steps of 0.1 (Section 13). A cell corner is 7.9 dB down at 1.4, so the scheduler must shift the lattice every frame.

9.3 Cumulative Pd of the search

A target closing at v is seen once per revisit RR, at ranges $R(1 + (k + \phi)q)$ with $q = vRR/R$ and a uniform phase ϕ; the look SNR scales as $(R/R_k)^4$. The requirement is that the *cumulative* probability

$$P_{\mathrm{cum}}(R) = \left\langle 1 - \textstyle\prod_k (1 - P_{\mathrm{d},k}) \right\rangle_{\phi} = 0.9$$

at the sector's required range (looks below $P_{\mathrm{d}} = 10^{-3}$ no longer count). Against a single look at R this changes the required SNR by $+6.0$ dB for $q = 1$ and -6.8 dB for $q = 0.1$.

9.4 False-alarm policy

By default a sector tolerates a number of false alarms per revisit (`fa_per_rr` $= 10$); the P_{fa} per cell is that number divided by positions times independent cells, bounded to $[10^{-14}, 10^{-4}]$ and rounded to a twentieth of a decade. Every false alarm is then confirmed in Ku like a detection. Ku looks use $P_{\mathrm{fa}} = 10^{-8}$.

10 Target models

Targets are described by *classes* whose values are *class means*, not data on a particular weapon: the adversary's equipment is unknown and every value of a plausible range is possible. A sector names a class and may override any value. ST is the conventional reference target of the benchmark; GENERIC completes a sector that names no class.

Class	v (km/s)	L (m)	ω (°/s)	$a_{\max}$	RCS S / Ku (m^2)	Fluct.	Depol.	T_0 S / Ku (s)
ST (reference)	5	10	0.1	0	1 000 / 1 000	SW1/2	−10 dB	3.1 / 0.6
GENERIC	10	3	1	1 g	1 / 1	SW1/2	−10 dB	1.0 / 0.20
KKV	25	1.5	1	10 g	1.27 / 13.8	SW3/4	−20 dB	2.0 / 0.41
SHIP	5	100	0.1	1 g	300 / 300	SW1/2	−6 dB	0.31 / 0.06
DEBRIS	15	0.5	10	0	0.01 / 0.01	SW1/2	−3 dB	0.61 / 0.12
DECOY	25	1.5	0.5	0	0.135 / 0.135	SW0	−20 dB	4.1 / 0.82

Frontal RCS and seeker optics. The RCS is co-polar and frontal (the aspect of a closing target). The KKV front is a large infrared seeker mirror, metallic and therefore a perfect conductor at radar frequencies, pointed at its target. Its contribution is $\sigma_{\text{opt}} = \max(\pi(2f)^2,\ \eta_r\, 4\pi A^2/\lambda^2)$ with $A = \pi D^2/4$, $D = 0.3\,\text{m}$, $f/D = 1$, $\eta_r = 0.1$: direct reflection on the concave primary, and the "cat's eye" retro-reflection of the focal assembly, in $1/\lambda^2$, hence mostly in Ku. The decoy mimics the body without the optics, which will help discrimination.

Fluctuation. The fluctuation of a look is a Gamma shape: the family (SW1/2: one degree of freedom per independent sample, SW3/4: two, SW0: steady) times $\min(n, 1{+}T_{\text{obs}}/T_0)$ independent samples over the look (n coherent blocks), times the polarisation diversity $(1+\rho)^2/(1+\rho^2)$ of the two receive channels. A declared Swerling case overrides the class.

Masking. The scene is unknown, so the masking of a weak target by a strong neighbour is an output: the envelope of the strong target's response at 1, 2 and 3 cells in range (with the look's weighting) and Doppler ($-13.3\,\text{dB}$ beyond one cell without a Doppler window).

11 Environment

11.1 Spacecraft structure

The obstacles are in the near field of the aperture (Fraunhofer limit about 325 m in S): the beam is a tube of the width of the aperture. Rays are cast from 37 points of the aperture along the pointing direction, exactly on the triangles of the 3D model; the blocked fraction f costs $-20\log_{10}(1-f)$ per pass, and a position beyond 3 dB per pass is masked. The structure also radiates: its emissivity times physical temperature (hull $\varepsilon = 0.01$ at 290 K, radiators 0.05 at 750 K) is picked up by the main lobe through the blocked fraction and by the far sidelobes as a body of the sky. With the 25° tilt, the structure only blocks directions more than 71.7° from a face normal, all served by another face at a smaller scan: no position is masked and the radiators are never seen; the hull adds a negligible 2.9 K.

11.2 Surfaces: shadow, clutter, converter saturation

Bodies with a surface (Earth, Moon) are spheres at a distance D given by their angular radius. A direction inside the disk meets the surface at $d = D\cos\delta - \sqrt{R^2 - D^2\sin^2\delta}$; nothing exists beyond. The surface backscatter is "constant-γ", $\sigma_0 = \gamma\cos\theta_i$ (γ = radar albedo; Earth $-12\,\text{dB}$ in S, Moon $-11.5\,\text{dB}$). The sidelobe clutter in the target's range cell r comes from the iso-range ring of the surface (area $2\pi R r\,\Delta r/D$ per cell), sampled on 720 points and seen through the two-way pattern: $y = \sigma_c/\sigma_{\text{target}}$. The main-lobe echo of a surface is checked against the ADC clipping level per element (alert), and a limb inside a sector is flagged (GO-CFAR advised there).

11.3 Clutter per waveform

In S, the clutter is evaluated *per candidate waveform and per 1° scan bin*, out of the T_{sys} budget (Ku keeps a simpler noise-factor model until its own step):

- **Clutter ratio** y, per pointing and per transmit beam of the search space (full face, or the 2D pattern of a partition sub-aperture), at the requirement range of the pointing, for a reference range cell; it scales exactly as $c/2B$ with the candidate's bandwidth, and is divided by the number of coherent blocks M (the per-block requirement).
- **Doppler residual** $\rho(v)$: the surface's Doppler spectrum (Gaussian, mean and 1-σ spread of its closing speed given by the scenario), folded over the PRF, through the filter of a coherent block (uniform weights w_k, or Taylor $-30\,\text{dB}$ and $-40\,\text{dB}$),

$$H(x) = \left|\textstyle\sum_k w_k e^{j2\pi kx}\right|^2 / \left(\textstyle\sum_k w_k\right)^2,$$

plus a white floor from the residual phase variance through the filter's noise bandwidth. $\rho = 1$ without coherent blocks or without the scenario's closing speed (prudent). It is evaluated at 11 target speeds over the sector's span.

- **Range**: clutter is treated as noise at the detector, $S/(N + C)$ with $C/N = a\,S/N$, $a = y\,\rho(v)/M$. In each bin y is taken at the 99 % quantile of the pointings; the SNR increase G that restores the look's P_d *averaged over the target speeds* is solved by bisection, and the range becomes $R_n G^{-1/4}$. If no power restores it, the candidate is unplayable in the bin; a pointing that no candidate can clear is blinded.
- **Requirement brought back to the surface**: where a pointing meets a surface at d below the required range, the requirement becomes $d -$ guard (guard = max(10 km, range resolution)), rounded *up* on a grid of 5 % of the range; the cumulative criterion is recomputed for each such class; with $d -$ guard ≤ 0 there is no volume and the pointing is not scanned.

11.4 Jammers

Stand-off jammers are declared in the sky. Their noise enters T_sys through the receive pattern: $T_j = (\text{EIRP density}/4\pi R^2)/k \times A_\mathrm{eff}(\text{direction})$. A jammer more than one receive beamwidth off axis is in the sidelobes and adaptively nulled by 40 dB (the 8 strongest per face).

Self-protection jammers are carried by the sector's target (barrage over the band of their class, EIRP densities from 5 dBW/MHz on a KKV to 27 dBW/MHz on a ship; in the main lobe, not nulled). They enter the range exactly: with $S \propto 1/R^4$ and $J \propto 1/R^2$, $S = \mathrm{SNR}_\mathrm{req}(N + J)$ is quadratic in $1/R^2$,

$$\frac{1}{R^2} = \frac{1}{2}\left(\frac{1}{R_{bt}^2} + \sqrt{\frac{1}{R_{bt}^4} + \frac{4}{R_n^4}}\right), \qquad R_{bt} = \sqrt{\frac{P\,G_t\,\sigma\,\tau N}{4\pi\,J_0\,\mathrm{SNR}_\mathrm{req}\,L}},$$

with R_n the range against noise alone and R_{bt} the burn-through range. **Concept of operation**: the search does not have to burn through a self-protection jammer; it identifies the jamming axis, and the scheduler then points a Ku burn-through search along it (Ku transmit gain about 25 times the S one, hence a burn-through range about 5 times larger for the same energy). Accordingly, a sector whose requirement the jammer puts out of the search's reach is planned against the noise alone and flagged for the Ku burn-through.

11.5 Other terms

Non-threatening objects (debris, satellites) seen per revisit are confirmed like targets. Faraday rotation over the round trip, $\Omega = 2\times 2.36\cdot 10^4\,B_\parallel\,\mathrm{TEC}/f^2$ (SI units), is reported from the scenario's electron content and field; it costs nothing with the dual-polarisation receiver.

12 Radar equation and loss budget

The range of one look is the energy form of the radar equation,

$$R^4 = \frac{P_\mathrm{beam}\,G_\mathrm{TX}\,G_\mathrm{RX}\,\lambda^2\,\sigma\,\tau N}{(4\pi)^3\,k\,T_\mathrm{sys}\,L\,\mathrm{SNR}_\mathrm{req}},$$

with $\mathrm{SNR}_\mathrm{req}$ the SNR on the total integrated energy from the exact detection law (cumulative in search), P_beam and G_TX those of the transmit mode, G_RX with the receive weighting of the volume, and T_sys the budget temperature of the direction. The total loss L is the sum of:

Fixed (S)	finite precision 0.3 dB, array errors after calibration 0.006 dB (Section 15), Ku-element blockage 0.2 dB, transmit chain 0.6 dB
Band	edge derating averaged over the occupied spectrum
Waveform	compression (weighting, NLFM mismatch, LFM Doppler), eclipsing of staggered trains, buffer droop, multibeam superposition, Doppler weighting
Antenna	active mismatch, transmit and receive
Polarisation	GLRT recombination; cross-polar energy of the target (a gain)
Processing	positions (grid cells, straddle, velocity-filter offset, through the averaged P_{d}), CFAR, coherence, acceleration, migration residual

For the reference waveform (24 pulses of 200 µs, 4-beam superposition) on the reference target at 2.8 GHz on boresight, the losses total about 6.6 dB, the position term included (Section C lists every term and its range); an independent hand derivation of the equation agrees within 0.5 % (self-test).

The cost of one search position is electrical: energy $P_{\mathrm{pk}}\, N\tau\, t/c_d/\eta$ and transmit time $N\tau\, t/c_d$, with t the trains per dwell (the beams of a time multiplex, 1 otherwise) and c_d the positions one dwell covers (1, B_{tx}, or B_{tx} times the sub-aperture cell ratio).

13 Search optimiser

For each sector, the optimiser (`core/optimizer.py`) chooses the S centre frequency and the waveforms, and computes the energy and time each face spends.

13.1 Frequency

The centre frequencies are *derived from the band*: the usable lower edge plus multiples of 5 MHz, kept where the signal fits inside the band and touches none of the spectrum exclusions of the mission. The search runs in two passes: the valid centres on a 20 MHz step (plus the centre-most one of every free fragment of the band), then the 5 MHz step within ±20 MHz of the best. The optimum is flat in frequency: the grid serves mainly to avoid jammed or forbidden spectrum. A sector may force its candidate frequencies.

13.2 Waveform plan

At each frequency, the ranges of all candidates are computed in every 1° scan bin. In the model the scan angle enters a range only through the gains and the active mismatch, so $R(\theta) = R(0)\, h(f, \theta)$ with h independent of the waveform (checked at run time): one solution per candidate and frequency suffices.

- One waveform per group of 5° bins, at most four distinct waveforms per sector; neighbouring groups share one by dynamic programming over contiguous segments.
- A candidate is admissible on a segment if it reaches the requirement in every populated bin. "No excess energy": where it exceeds the requirement, the *pulse is shortened* to $s\tau$ with $s = (R_{\mathrm{req}}/R)^4$ (PAs stay saturated; PAE of the shortened pulse exact; floor $\tau B \geq 1$; the losses of the full pulse are kept, which is prudent). Energy and transmit time both decrease.
- Objective: minimise max(energy fraction, transmit-time fraction) over the faces, by a weighted sum whose weight is swept (21 values).
- With no admissible candidate on a segment: the fewest positions short of range, then the longest reach, then the cost. A sector whose requirement a self-protection jammer puts out of reach is planned against the noise alone (Section 11).

- Ranking of frequencies: the unscanned fraction caused by the frequency first, then the positions short of range (1 % tolerance), then the objective, then a deterministic secondary criterion (energy, time, number of waveforms, distance to the core-band centre).

Dominated candidates (same cost function, range lower or equal everywhere) are pruned exactly before the plan. Every search lattice spacing (1.3 to 2.0 transmit beamwidths) is evaluated at every centre frequency, and the cheapest plan is kept: the spacing changes the positions, the grid losses, the receive beams per position, the false-alarm cells and the Ku budget that follow. When a face is over-subscribed, the revisit is stretched in proportion.

13.3 Outputs and flags

Per sector: frequency and occupied band, waveforms per scan bin with their timing, ranges at best and worst scan, positions, receive beams per position, sky and environment summaries, energy and time per face, revisit alone and in the mission. Flags: resolution not met, range not reached, power or time over-subscribed, unscanned fraction, active mismatch, receive sidelobe option not held, sky blinding or rise above the budget, masking, clutter limitation, surfaces closer than the requirement, self-protection jamming, revisit below three target coherence times (the cumulative criterion assumes independent looks), spectrum exclusions removing most centres.

14 Confirmation and active tracking in Ku

Every S detection, true target or false alarm, is confirmed in Ku (`core/tracking.py`, roadmap step 3.2). Three modes:

- **KU1+7**: one Ku look on the S estimate, then a cluster of 7 beams on failure (two round trips);
- **KU7**: the cluster of 7 beams at once (one round trip);
- **S2**: a pointed S "touch" (an S waveform of role CONFIRM, which also rejects the S false alarms), then KU1+7 on the improved angle estimate.

The cluster is sent as 7 successive looks, as a phase-only superposition of the 7 beams (power / 7, loss interpolated in the multibeam table, 0.76 dB), or by a partition of the face into 2 or 4 Ku sub-apertures; the cheapest is kept. The echoes that cross the threshold have a random SNR, drawn by quantiles of the target's Gamma law; the angle error per axis is $K\,\mathrm{HPBW}/\sqrt{\mathrm{SNR}}$ (Cramér–Rao bound $\times 1.13$) plus a residual S/Ku alignment bias, averaged by Gauss–Hermite nodes. Each mode gets the smallest power margin that confirms a true target with probability at least 0.98, solved by bisection; the mode that minimises the load of the most loaded face is chosen, or the operator imposes one.

The Ku frequency is chosen per sector on a 10 MHz grid derived from the Ku band, the signal inside the band and outside the mission's Ku exclusions; the S fifth harmonic on the Ku channel of the same face is flagged as a scheduler time rule. Ku looks are chosen in their own space (pulses 5 µs to 2.08 ms, PRIs 0.5 ms to 5 ms, bandwidths 25 MHz to 200 MHz, both integrations, an agile variant of noncoherent trains), shortened at saturated PAs to just reach the range, with the sector's Ku sky budget, its self-protection jammer and the sidelobe clutter of its surfaces; a Ku burn-through look is pointed in the jamming axis when the jammer puts the sector out of search reach.

Active tracking is described per range bracket (number of tracks, update period wanted by the operator, angular zone, range and radial-velocity accuracies) and follows a track model (`core/trackmodel.py`, roadmap step 3.3). Each axis (range; two cross-range axes) carries a steady-state Kalman filter with three states and the Singer manoeuvre model ($\sigma_m^2 = a_{\max}^2/3\,(1+4p_{\max}-p_0)$ plus a floor, time constant per class), discretised exactly and solved by the discrete Riccati equation. The angular zone (σ = zone / 4) applies to the predicted angle at the next update, the range and velocity

accuracies to the filtered track; each requirement becomes the largest measurement error that meets it, measurement floors (0.05 mrad, 0.05 m, 0.01 m/s) taken out in quadrature, and the range accuracy pays the range–Doppler coupling residual $c\tau\sigma_{f_d}/2B$. The update look is pointed at the prediction: its pointing error enters its P_d as a loss law on the real Ku pattern; it banks over $\pm 3\sigma$ of the predicted radial velocity and acceleration; the cheaper of one pointed look and a 7-beam sequential cluster is kept. A Markov chain over consecutive misses (track lost after 3, the pointing error growing while coasting) gives the loss rate; initiation looks bring a confirmed target to the required accuracies; costs are averaged over 8 ranges of the bracket. When the operator's period cannot be held at some range, the longest tenable period is reported as an alert. A KKV is told from a decoy on the track looks through four indices (depolarisation, fluctuation, RCS level with the seeker optics, manoeuvre seen by the filter), Faraday uncertainty weakening the depolarisation index unless the transmit polarisation is switched. A post-LNA attenuator (30 dB) lowers the range below which the echo clips the ADCs by a factor 5.6. S and Ku share the electrical energy of a face; their transmit times are counted separately.

Calculation settings. Since 25 October 2026 approximations are allowed when they do not denature the model, each a `[tool.*]` entry whose exact value stays available: position-loss laws rounded to 0.02 dB (required SNR within 0.003 dB), cumulative inversion to 1×10^{-3} dB, lattice spacing searched at the frequency kept, three S and three Ku centres over the core band when the spectrum is quiet (no exclusion, no jammer), S2 margins to 0.05 dB with 16 SNR quantiles and 8 Gauss–Hermite nodes. On two test missions they change the search energy by at most 0.25 % (one sector 9 %, one −5 %: the frequency matters for some sectors) for a 7 to 8 times faster run.

15 Other functions

Array errors and calibration. Every TRM is calibrated in flight through its coupler (transmit and receive loopback, all TRMs of a face at once with orthogonal codes), 10 ms in S and 50 ms in Ku every 300 s, taken from the face's transmit time. The residual errors at the end of an interval (random per TRM 1° / 0.1 dB, correlated per TRSA 0.5° / 0.05 dB plus a thermal drift of 0.1 °/min / 0.01 dB/min) give the gain loss by Ruze's formula on transmit and receive (about 0.006 dB, replacing a lumped 0.5 dB / 0.8 dB), a mean sidelobe floor near −59 dB and a boresight error of a few hundredths of a milliradian; the angle-measurement floor and the S/Ku alignment bias stay inputs, checked against them. Degraded modes and TRM failures are left to the end of the development (nominal operation first).

Redundancy. A sector A covers the need of a sector B if, with A's waveforms and revisit, B's target, range and receive weighting (the digital receiver can apply both on the same look), the cumulative criterion of B is met, with an unambiguous PRI and sufficient resolution.

Fast estimate. For live feedback while the operator edits, a precomputed table gives the cheapest energy per position versus range, RCS, speed and scan angle; only the full calculation is authoritative.

Frames and navigation. The navigation message carries a Hamilton quaternion (ship to ICRF) and an optional orbit. Before any calculation, the mission is turned into the ship frame once: each sector given in another frame (inertial, RTN, limb-relative, two directions) receives its rotation matrix and the direction raster is rotated before the shape test; the inertial sky is rotated back. All later calculations stay in the ship frame.

16 Software

16.1 Modules

Module	Role
`core/params.py`	loads `radar_system.toml` once per process; an alternative file via `RADAR_PARAMS`
`core/radar_model.py`	bands, TRMs, patterns, losses, waveforms, processing, targets, `solve()`
`core/antenna.py`	radiation patterns from the build-up, far-lobe profiles, sub-apertures (cached)
`core/detection.py`	exact detection laws, averaged and cumulative P_{d}
`core/sky.py`	sky noise per direction, blinding, sector budget
`core/environment.py`	structure (3D model, ray casting, cached), surfaces and clutter, Doppler residual, jammers, Faraday
`core/isolation.py`	leakage between bands and faces, scheduler rules
`core/frames.py`	navigation, frames of the volumes
`core/optimizer.py`	S search optimiser, mission run, sector summaries
`core/tracking.py`	Ku confirmation and active tracking budget
`core/trackmodel.py`	track model: Singer Kalman filter, loss chain, initiation, discrimination
`core/redundancy.py`	redundancy between sectors
`core/cost_estimate.py`	fast estimate for the interface
`core/mission.py`	mission model, validation, JSON / CSV
`core/geometry.py`	sphere raster, 3D view of the ship
`gui/`	PyQt5 interface with 3D views (PyVista)

Python 3.11 or later, NumPy, and SciPy for the track model (discrete Riccati equation, matrix exponential; the Bessel functions of the patterns are coded directly); Numba, when installed, compiles the sums of the body noise (same formulas in double precision, parallel; the NumPy path gives the same result to 10^{-14}). The optimiser runs sectors, and the frequencies of a sector with all their lattice spacings, in worker processes; the detection laws, position-loss laws and cumulative-P_{d} inversions are cached exactly.

16.2 Tests

The test bench has two families: frozen outputs of every test mission compared at a relative tolerance of 10^{-9} (non-regression), and physical laws and invariants that must hold whatever the values ($R \propto \sigma^{1/4}$, exactness of every shortcut, consistency of caches, detection laws against closed forms, ...). Every refreeze of the frozen outputs is recorded with its cause. An impact report lists the key figures and the waveforms that change before any refreeze. At model V6.16.0 the bench has 205 tests (11 of them compare the frozen outputs of the test cases), run at three levels: fast tests, tests touched by a change, and the full pass (cost table, every test case computed once and kept with a fingerprint of the code, impact report, whole bench; about 19 minutes, with a global progress bar).

17 Roadmap and open points

Phase	Steps (status on 25 October 2026)
0, foundations	single parameter file, test bench, review of every value — done
1, antenna and RF	real pattern; active mismatch, grating and gap lobes; TRMs and transmit chain; band; receive chain, sky noise, converters, isolation — done
2, signal and detection	parametric waveforms; processing; detection criterion; target models; environment — done
3, radar functions	3.0 frames and navigation; 3.1 S search; 3.2 Ku confirmation; 3.3 tracking; 3.4 array errors and calibration — done (degraded modes and TRM failures deferred to the end of the development)
4, resources	4.1 power: shared ship budget, always-on consumption, TRM supply, face storage — coded; thermal; scheduler and latency; computing and data
5, mission and validation	richer mission model; global optimiser; validation engine; Monte-Carlo simulation; operator interface; verification and documentation

Main provisional dispositions, each assigned to a later step:

- **Scheduler (4.3)**: common cadence of the faces per band and the S-harmonic rule; search-lattice shift every frame; eclipsing by other looks and faces; latency rebuilt from scratch; time multiplex and superposition re-evaluated once a timeline cost exists; the Ku burn-through search pointed along a jamming axis; choice of the CFAR per pointing; longest contiguous transmission allowed by the face storage; S/Ku alignment on external references.
- **Thermal (4.2)**: micro-channel capacity and time constants; pumps; the drift that sets the calibration interval.
- **Computing (4.4)**: simultaneous receive beams, converter data rates, beamforming power, computing load of velocity and acceleration banks and of adaptive nulling.
- **Mission and validation (5.x)**: per-class RCS spread and RCS versus aspect (discrimination is optimistic today), track-update and discrimination probabilities as mission settings, joint optimisation of the power split among faces, recalibration of the fast estimate on the track model.
- **Antenna**: element pattern and mutual coupling at tile edges not modelled (isotropic elements); couplings between faces are optimistic orders of magnitude needing an electromagnetic simulation.
- **End of development**: degraded modes (face, TRMs, power or Ku aperture lost) and TRM failures over a mission.

A Key derived values

Quantity	Value
Search positions per face at 2.8 GHz	about 1 664 over the full sector (lattice of 1.4 transmit beamwidths)
Receive beams per position	6.1 / 5.3 / 4.8 at −25 dB, −30 dB and −35 dB (spacing 0.5 receive beamwidth)
Mean transmit grid loss over a cell	3.4 dB (real pattern); receive grid 0.40 dB
Transmit power available per face (reference 2 MW)	1.784 MW electrical (chain 0.945, always-on 106.9 kW)
Peak electrical power of a full-power look	12.3 MW S, 14.1 MW Ku (face storage 5 MJ)
Longest pulse	2.0 ms in S (300 km blind), 2.08 ms in Ku
PAE at $d = 0.65$, long pulse	0.778 in S, 0.621 in Ku
ADC noise figure	20.0 dB (13 bits, 10 dBm, 250 MS/s)
Noise at the ADC	−56.4 dBFS in S
Minimum unsaturated range (S, boresight)	reference target 12.2 km, KKV 2.3 km, ship 9.1 km
Range weighting −25 dB, −30 dB and −35 dB	loss 0.41 dB, 0.68 dB and 0.93 dB, widening ×1.18 / 1.27 / 1.34
Doppler at 25 km/s	470 kHz in S; NLFM possible above 23 MHz in S, 117 MHz in Ku
Round-trip active mismatch at 2.8 GHz	0.26 dB at 0°, 0 at 45°, 1.12 dB at 70°, 2.11 dB at 75°
Sector area in (u, v)	2.0006

B Glossary: quantities and the physics behind them

This appendix explains, for each main quantity of the model, what it means and which physical phenomenon it stands for. Values are those of the parameter file at the date of the document.

B.1 Geometry, antenna and beams

Quantity	Meaning and physics
λ, f	Wavelength and carrier frequency, $\lambda = c/f$ (10.7 cm at 2.8 GHz, 2.14 cm at 14 GHz). Every angular size of the array scales as λ/D.
D_{env}, x_{rms}	Envelope of the aperture (4.17 m) and RMS extent of the element positions along one axis. λ/D_{env} sets the beamwidth; x_{rms} sets the best possible angle accuracy (an aperture measures the phase slope of the wavefront across its extent).
(u, v), θ	Direction cosines in the face plane and scan angle off the normal. A planar array forms a beam of constant width in (u, v); in angle it widens as $1/\cos\theta$ because the projected aperture shrinks.
HPBW	Half-power beamwidth: the angle between the two directions where the gain is halved. Transmit (uniform illumination) $1.036\,\lambda/D_{\text{env}}$; receive wider because of the weighting.
G, η	Gain $4\pi A\eta\cos\theta/\lambda^2$: how much the array concentrates power in the beam direction compared with an isotropic radiator. η is the illumination efficiency: 1 for uniform excitation, less with a taper that lowers the sidelobes (energy taken from the edges).

Quantity	Meaning and physics
Sidelobe level	Relative gain outside the main lobe. Low sidelobes reject strong sources off the axis (Sun, Earth, jammers, clutter) at the cost of gain. The receive weighting (Taylor law) sets it per volume.
Grating lobe	A full replica of the main lobe that appears when the element spacing exceeds about $\lambda/(1+\sin\theta)$: it sets the top of the usable band and the scan limit.
Active mismatch Γ	In a scanned array the elements couple to one another; the impedance seen by each TRM changes with the scan, part of the power is reflected. The WAIM layer matches the array at 45°; the loss rises near the grating-lobe onset.
TRSA, gaps	Transmit–receive sub-array tile and the dead strips between tiles. The break of the regular spacing at the gaps creates "gap lobes" that weighting cannot lower.
Lattice spacing	Distance between neighbouring search beams, in transmit beamwidths (1.3 to 2.0). Wider spacing means fewer positions but a target further from the nearest beam centre (grid loss).

B.2 Transmitter and power

Quantity	Meaning and physics
P_{pk}	Peak RF power of one TRM at the PA output (2 kW S, 75 W Ku). The PAs run saturated (constant envelope), the most efficient regime.
PAE η	Power-added efficiency: RF power out over DC power in. Lowered by the pulse edges (DC drawn while the RF settles) and by channel heating at high duty.
τ, PRI, d	Pulse length, pulse repetition interval and duty cycle $d = \tau/\mathrm{PRI}$. Long pulses carry more energy (range) but blind the receiver longer (blind range $c\tau/2$).
N, train	Number of pulses in one look; their echoes are integrated coherently (phase kept) or noncoherently (powers added).
Energy buffer	Local capacitor bank of each TRM that delivers the pulse; the voltage sags as it empties (droop) and the PA output follows.
TRM supply	Continuous power that refills the buffer between pulses; it bounds the mean power of a train.
Chain efficiency	Power supply and distribution efficiency (0.945): the share of the bus power that reaches the TRMs and the receivers.
Always-on power	Receivers, converters, beamforming, control: consumed whether the face transmits or not.
Face storage	Energy reserve that delivers the peak power of a look (all TRMs at once) above what the bus supplies on average.

B.3 Receiver, noise and interference

Quantity	Meaning and physics
T_{sys}	System noise temperature: the thermal noise power $kT_{\mathrm{sys}}B$ the receiver adds, referred to the antenna. It sums the sky seen by the antenna, the losses before the first amplifier (a lossy part at temperature T radiates $(L-1)T$), the amplifier noise and the converter noise, each divided by the gain before it (Friis).
Noise figure, ENOB	Noise figure: how much a stage degrades the SNR. ENOB: effective number of bits of the converter; its quantisation noise gives the converter's noise figure.
Clipping, dynamic range	A converter saturates above its full scale; since each element is digitised before beamforming, a strong echo or a bright surface in the main lobe can clip all converters at once.

Quantity	Meaning and physics
Sky noise T_b	Every body emits thermal noise at its brightness temperature (Sun 1×10^5 K in S, Earth 290 K, Moon 230 K, galaxy a few K). Seen through the antenna pattern it raises T_{sys}; a body in the main lobe can blind a direction.
Polarisation, ρ	The echo's polarisation is unknown; two receive channels collect both components. ρ is the target's cross-polar to co-polar RCS ratio: the extra energy the second channel collects.
Faraday rotation	The ionosphere's electrons in the magnetic field rotate the polarisation plane by $\propto B_{\parallel}\,\text{TEC}/f^2$; harmless for detection with two channels, a source of error for polarimetry.
Clutter	Echoes of a surface (Earth, Moon) at the same range as the target, through the sidelobes, competing with it. Doppler filtering rejects it when the speeds differ.
Jammer	Noise sent toward the radar: stand-off (in the sidelobes, nulled adaptively) or self-protection (carried by the target, in the main lobe). Burn-through range: where the echo again beats the jamming.

B.4 Waveform, processing and detection

Quantity	Meaning and physics
B, pulse compression	Signal bandwidth. A long coded pulse compressed on reception has the energy of the long pulse and the range resolution $c/2B$ of a short one. LFM: linear frequency sweep, Doppler tolerant; NLFM: non-linear sweep whose spectrum shape gives low range sidelobes without weighting loss.
Range sidelobes, weighting	Residual response of the compressed pulse beside its peak; a weighting lowers them (to keep a weak target visible next to a strong one) at the cost of SNR and resolution.
Doppler f_d	Frequency shift $2v/\lambda$ of the echo of a target closing at v (470 kHz at 25 km/s in S). It shifts the compressed peak (coupling) and separates targets from clutter.
Coherent block, T_0	Part of a train integrated with its phases. It works while the target's echo phase stays predictable: a rotating target decorrelates in its coherence time $T_0 = \lambda/(2L\omega)$, an oscillator's phase noise also spoils it.
Range migration	A fast target moves through several range cells during the look; it is compensated, at the cost of a bank of velocity hypotheses.
Velocity, acceleration banks	The processing tries several speed (and acceleration) hypotheses; each costs a filter (computing) and counts as extra cells for false alarms; a target between two hypotheses loses SNR.
Straddle	A target falling between two samples of the range or Doppler grid loses SNR; oversampling reduces it.
CFAR	Constant false-alarm rate detector: the threshold is set from the noise measured in nearby cells; the estimate's randomness costs a small loss.
P_{d}, P_{fa}	Probability of detecting a target and of a false alarm per independent cell.
Swerling, Gamma shape k	Statistical model of the target's echo power from look to look: steady (sphere), Swerling 1/2 (many scatterers, exponential), Swerling 3/4 (one dominant scatterer). The model uses a Gamma law of shape k: the larger k, the steadier the echo; independent samples (pulses, frequencies, polarisations) raise k.
SNR_{req}	SNR the detector needs to reach the required P_{d} at the given P_{fa} and fluctuation.
Cumulative P_{d}, q	Probability that a closing target has been detected at least once by the time it reaches the required range; $q = vRR/R$ is the relative distance it covers between two looks.
Positions loss	SNR lost because the target is somewhere inside the search cell (beam centre, range and Doppler grids, velocity filter), averaged in P_{d}.

B.5 Confirmation and tracking

Quantity	Meaning and physics
Confirmation	A Ku look aimed at an S detection: the narrow Ku beam (5 times narrower) checks it is real before a track is opened; the angle error of the S estimate decides whether one beam or a cluster of 7 is needed.
Kalman filter, Singer model	A recursive estimator of position, velocity and acceleration. The Singer model treats the target's acceleration as a random process of standard deviation σ_m and time constant τ_m (how long a manoeuvre lasts).
Predicted, filtered error	Uncertainty on the target just before an update (where to point) and just after it (track quality delivered).
Pointing loss	The update beam is aimed at the prediction; the target is off the beam centre by the prediction error, losing gain on transmit and receive.
Coasting, track loss	An update missed (no detection) leaves the track to extrapolate one more period; the error grows, the next look is harder; a few misses in a row lose the track.
Discrimination	Telling a warhead (KKV) from a decoy: polarisation (depolarisation ratio), fluctuation statistics, RCS level (seeker optics) and manoeuvres differ.
Array errors, calibration	Residual amplitude and phase errors of the TRMs after calibration: they lower the gain (Ruze) and raise a floor of random sidelobes; they drift with temperature between calibrations.

C Modelled effects and their impact

Every effect the model accounts for, what it represents, and the range of its impact. Unless stated otherwise the impact is an SNR loss in dB (a gain when negative): a loss of x dB shortens the range by the factor $10^{-x/40}$ (1 dB $\approx -5.6\,\%$, 3 dB $\approx -16\,\%$) or, at a fixed range, multiplies the energy by $10^{x/10}$. The ranges come from the parameter file and from a sample of the S search space (3 test sectors, 900 waveforms, 3 frequencies, 0 to 70° scan) and of the Ku look space (300 looks); they are test values, not performances.

Effect	What it represents	Impact (range)
Antenna and transmit chain		
Projected aperture $\cos\theta$	The array seen from the scan direction is smaller: gain on transmit and receive	0 to 9.3 dB two-way at 70°
Active mismatch	Reflection of the TRM power by the array's coupled impedance, scan dependent, WAIM matched at 45°, rising near grating lobes	0 (45°) to 1.1 dB at 70°, 2.1 dB at 75° (2.8 GHz, two-way)
Transmit chain	Harmonic filter, circulator, coupler, line between PA and element	0.60 dB S, 0.95 dB Ku
Blockage by the other band	Elements of the other band scatter part of the field	0.2 dB S, 0.3 dB Ku
Array errors (calibration)	Residual TRM amplitude and phase errors and thermal drift between calibrations (Ruze)	0.006 dB; sidelobe floor −59 dB
Receive weighting option	Taylor weighting at −25 dB, −30 dB and −35 dB sidelobes	gain 0, −0.53, −1.00 dB
Band edges	Raised-cosine derating of the TRM outside the core band, averaged over the occupied spectrum	0 (core) to 1.0 dB S / 0.5 dB Ku at the edges (sample: 0–0.17)
Energy buffer droop	PA output following the sagging buffer voltage during the pulse	0 to 0.46 dB
PAE (edges, duty)	DC power drawn without full RF on pulse edges, channel heating at high duty	energy +0.1 to +2.8 % S, up to +4.4 % Ku

Effect	What it represents	Impact (range)
Transmit multibeam	Phase-only superposition intermodulation; partition: smaller gain, wider beam	0.4 / 0.6 / 0.8 dB for 2 / 4 / 8 beams
Waveform and processing		
Pulse compression	NLFM mismatch, or LFM weighting plus Doppler loss	0.1 dB (NLFM); LFM 0.41 to 0.93 dB + Doppler (Ku sample up to 1.26 dB)
Doppler weighting	Taylor weighting of a coherent block against clutter	0.68 dB (−30 dB), 1.14 dB (−40 dB)
Eclipsing	Echoes lost while the receiver is deaf during later pulses of a range-ambiguous train	$-10\log_{10}(1-(\tau+t_{\mathrm{rec}})/\mathrm{PRI})$, typically 0.1 to 1 dB
Coherence	Target decorrelation and oscillator phase noise inside a coherent block	0 to 0.63 dB
Range migration	Residual of the compensation when the target crosses several cells	0 or 0.1 dB
Velocity bank	Target between two velocity hypotheses (mean over the step)	0 to 0.34 dB
Acceleration	Quadratic phase and range curvature of an accelerating target, or bank residual	0 to 0.35 dB
Range and Doppler straddle	Target between two grid samples (grids oversampled ×4)	0.05 dB and 0.07 dB
CFAR	Randomness of the noise estimate (CA, 1024 cells; GO at clutter edges)	0.04 dB (GO 0.05 dB)
Finite precision	Quantisation and arithmetic of the digital processing	0.3 dB
Detection		
Search grid (transmit)	Target anywhere in the hexagonal cell of the search lattice, real pattern	3.3 dB mean (spacing 1.4), more for wider spacings
Receive beam lattice	Target between the digital receive beams (spacing 0.5 receive beamwidth)	0.40 dB
Positions loss (total)	All the above position effects through the P_{d} averaged over the cell	3.1 to 4.3 dB (search), 0.05 to 0.5 dB (Ku looks)
Target fluctuation	Echo power varying from look to look; steady, Swerling 1, Swerling 3 at $P_{\mathrm{d}} = 0.9$, $P_{\mathrm{fa}} = 10^{-8}$, one sample	$\mathrm{SNR}_{\mathrm{req}}$ 14.2 / 22.4 / 18.5 dB (fluctuation costs up to 8 dB; agility and diversity recover part of it)
False-alarm policy	P_{fa} per cell from the tolerated false alarms, 10^{-14} to 10^{-4}	$\mathrm{SNR}_{\mathrm{req}}$ ±3 dB around 10^{-8}
Cumulative criterion	Several looks at a closing target instead of one	−6.8 dB ($q = 0.1$) to +6.0 dB ($q = 1$) against one look
Polarisation (GLRT)	Unknown echo polarisation estimated over the look	0.03 to 0.47 dB
Cross-polar energy	Second channel collects the cross-polar echo	gain −0.04 (KKV) to −1.76 dB (debris)
Receiver and environment		
System noise (cold sky)	Friis cascade: sky, pre-LNA losses, LNA, post-LNA, converter	T_{sys} 144.5 K S, 230.8 K Ku
Galactic background	Galaxy brightness through the pattern (worst case without a frame)	up to +8 K (≈ +0.2 dB in S)

Effect	What it represents	Impact (range)
Bodies (Sun, Earth, Moon)	Brightness temperature through the receive pattern	Earth +2.5 to 4 dB S; Moon +2.6 dB Ku; Sun ≈ +17 dB: blinded above 10 dB
Spacecraft structure	Blockage of the beam tube by the hull; emission of hull and radiators	0 with the 25° tilt; hull +2.9 K
Surface clutter	Sidelobe echoes of Earth or Moon in the target's range cell, after Doppler filtering	0 to blinding (SNR increase G per waveform); requirement brought back to the surface
Stand-off jammers	Noise through the sidelobes, adaptively nulled (40 dB)	T_{sys} rise, direction dependent
Self-protection jammer	Barrage noise in the main lobe	range limited to the burn-through range (Ku burn-through look in the jamming axis)
Faraday rotation	Polarisation rotation by the ionosphere	0 for detection; error on polarimetry
Converter clipping	Strong echo or bright surface saturating every element's converter	minimum unsaturated range (S KKV 2.3 km); post-LNA attenuator lowers it by up to ×5.6
Blind range	Receiver deaf during the pulse and limiter recovery	$c(\tau + t_{\mathrm{rec}})/2$: 3 km (20 µs) to 300 km (2 ms)
Confirmation and tracking		
S-to-Ku angle error	S estimate error and S/Ku bias against the narrow Ku beam	confirmation margins 0 dB to 12 dB, single look or cluster of 7
Track pointing loss	Prediction error of the track against the Ku beam (transmit and receive)	0.4 (0.1 beamwidth) to 3.9 dB (0.3 beamwidth); cluster cell 3.8 dB on 7 looks
Track banks	Velocity and acceleration hypotheses over $\pm 3\sigma$ of the prediction	0 to ≈0.4 dB
Range–Doppler coupling	Residual of the coupling after Doppler correction, on the range accuracy	raises the SNR of range-driven updates (long pulses, narrow band)
Track loss and initiation	Missed updates, coasting, re-initiation; initiation looks after confirmation	loss 10^{-3} to 10^{-2} per update; initiation 1 to ≈ 30 looks
Measurement floors	Calibration, TRM errors, clock, oscillator	0.05 mrad, 0.05 m, 0.01 m/s
Discrimination	Four indices on the track looks	looks needed to reach $P = 0.95$ (optimistic without class spread)
Resources		
Calibration time	In-flight calibration taken from the transmit time	3.3×10^{-5} (S), 1.7×10^{-4} (Ku) of the time
Always-on power	Converters, receive chains, beamforming, control	106.9 kW per face (≈ 5 % of 2 MW); pumps 2 % of the heat
TRM supply	Mean power and buffer recharge over a train	removes waveforms (0 in S, 3 % of Ku looks)

Effect	What it represents	Impact (range)
Face storage	Peak power of a look above the bus mean	scheduler rule $\approx$0.3 s of contiguous transmission
Calculation settings	Approximations of the optimiser (rounded laws, few frequencies in a quiet spectrum, . . .)	$\leq$0.003 dB on $\mathrm{SNR}_{\mathrm{req}}$; search energy -5 to $+9$ % per sector